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ORIGINAL ARTICLE

Length of surgery and pressure ulcers risk in cardiovascular surgical patients: a dose–response meta-analysis

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Key words
Cardiovascular surgery; Dose–response relationship; Length of surgery; Meta-analysis; Pressure ulcers

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Abstract

The aim of this study was to assess the relationship between length of surgery (LOS) and pressure ulcer (PU) risk in cardiovascular surgery patients. PubMed and Web of Science were systematically searched. We compared LOS difference between PU (+) group and PU (-) group. We also examined the dose–response effect of this relationship. The mean LOS in the PU(+) groups ranged from 252·5 to 335·7 minutes, compared with 233·0 to 298·3 minutes in PU(−) groups. The LOS was higher in PU(+) groups compared with PU(−) groups [weighted mean difference (WMD) = 36·081 minutes; 95% CI: 21·640–50·522 minutes; $Z = 4{\cdot}90$, $P = 0{\cdot}000$]. The funnel plot showed no publication bias. A significant dose–response association was also found between the LOS and the risk of surgery-related pressure ulcers (SRPU, model $\chi^2 = 9{\cdot}29$, $P = 0{\cdot}000$). In the linear model, the PU OR was 1·296 (95% CI 1·097–1·531) for a 60-minute increase in the LOS intervals and 13·344 (95% CI 2·521–70·636) for a 600-minute increase. In a spline model, the OR of PU increased almost linearly along with the LOS. Our meta-analysis indicated that LOS was an important risk factor for pressure ulcers in cardiovascular surgical patients.

Introduction

Cardiovascular surgery patients are among those most at risk of developing surgery-related pressure ulcers (SRPU), with a reported incidence as high as 29·5% (1). Pressure ulcers not only make a significant independent contribution to the excess length of hospitalisation (2) but also increase significant additional costs for pressure ulcer treatment (3). Most importantly, pressure ulcers are associated with a higher risk of mortality (4). Many studies have examined factors associated with SRPU development in cardiovascular surgery patients. Commonly cited risk factors include temperature manipulation, vasoactive drugs, hypotensive periods and reduced haemoglobin and haematocrit levels (5).

A pressure ulcer is a localised injury as a result of pressure or pressure in combination with shear and/or friction. Cardiovascular surgery is always related with prolonged surgery, which increases the pressure on local tissue and increases the risk of SRPU. There have been some studies investigating the relationship between length of surgery (LOS) and the SRPU risk in cardiovascular surgery patients. Some studies found that LOS was an important risk factor for SRPU (6), but other studies found there was no significant difference of LOS between SRPUs developed and no SRPUs developed in cardiovascular surgery patients (*P*-value was 0·34 in Stordeur's study (7) and 0·43 in Jesurum's study (8)). We found that there is still no systematic review summarising the relationship between LOS and SRPU risk in cardiovascular surgery patients.

The aim of this study was to assess this relationship by meta-analysis and find the dose–response effect of this relationship.

Key Messages

- the mean length of surgery is longer in pressure ulcer (+) group compared with pressure ulcer (+) group
- there is a significant dose–response association between length of surgery and risk of pressure ulcer
- pressure ulcer risk increases almost linearly along with length of surgery

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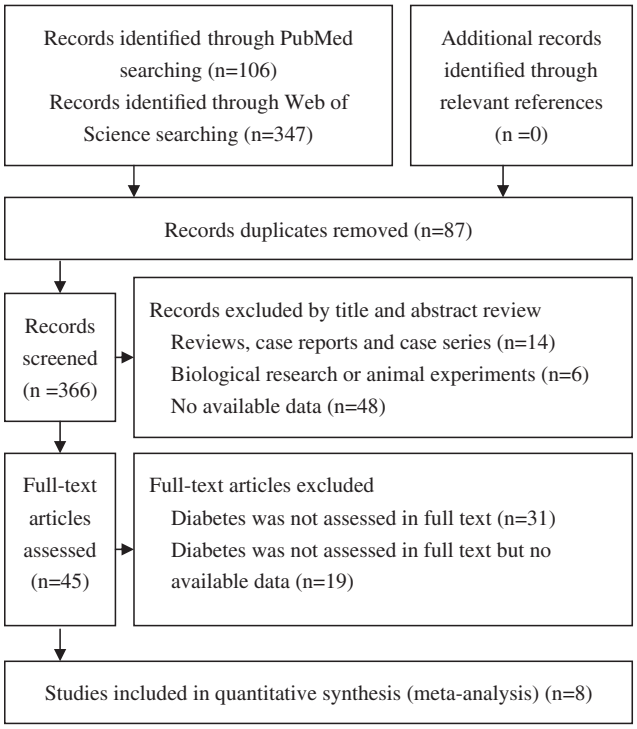

Figure 1 Flow diagram showing selection of studies.

Methods

Databases and literature search

We searched MEDLINE and Web of Science up to 20 September 2016. The search terms included 'pressure ulcer', 'cardiac surgery', 'cardiovascular surgery' and 'cardiothoracic surgery'. The search detail in MEDLINE was 'pressure ulcer'[MeSH Terms] AND [cardiac (Title/abstract) OR cardiovascular (Title/abstract) OR cardiothoracic (Title/abstract)]. In Web of Science, the search detail used was as follows: (Ts = pressure ulcer*) AND (Ts = cardiac OR Ts = cardiovascular OR Ts = cardiothoracic). We supplemented our searches by manually reviewing the references of all relevant studies.

Eligibility criteria

The following eligibility criteria had to be fulfilled: (i) observational studies (cross-sectional, case–control or cohort study) that permitted assessment of LOS and SRPU risk; (ii) outcome was the incidence of SRPU during the cardiovascular perioperative period; (iii) study reported the sample size, mean and standard deviation of LOS both in the SRPU group and the no SRPU group; and (iv) study divided LOS into three or more quantitative categorised levels and reported the number of cases and controls for each LOS category. The eligibility criteria was described as: (i) AND (ii) AND ((iii) OR (iv)).

Data extraction and critical appraisal

The following data were abstracted onto standardised forms: first author; publication year; country; study design; enrolment

Table 1 Characteristics of the included studies accessing length of surgery and pressure ulcer risk in cardiovascular surgical patients

First author, year	Country	Study design	Enrolment year	Sample size	Age (year)	Gender (M/F)	SRPU definition (stage)	SRPU incidence	SRPU(+) group		SRPU(-) group		NOS
									N	LOS (min)	*N*	LOS (min)	
Lu, 2016 (13)	China	Prospective cohort	2015	149	49·8 ± 17·7	79/70	I–IV	24·8%	37	263·6 ± 93·0	112	221·7 ± 85·8	8
Chen, 2015(14)	China	Retrospective cohort	2012	286	46·9 ± 22·1	160/126	I–IV	16·4%	45	259·7 ± 108·9	230	182·6 ± 98·8	8
Schuurman, 2009 (15)	Netherlands	Prospective cohort	2000	204	68·1 ± 9·6	129/75	I–IV	57·4%	117	252·5 ± 89·6	87	233·0 ± 81·2	7
Stordeur, 1998 (7)	Belgium	Cross-sectional	1995	163	64·5 ± 11·3	NR	II–IV	29·4%	48	283·8 ± 122·1	115	265·1 ± 91·8	7
Jesurum, 1996 (8)	USA	Prospective cohort	1995- 1996	36	63 ± 10·7	26/10	I–IV	16·7%	6	335 ± 167·4	30	280 ± 81·2	7
Papantonio, 1994 (16)	USA	Cross-sectional	1994	136	61·9 ± 13·4	90/46	I–IV	27·2%	36	335·7 ± 120·9	97	298·3 ± 85·2	7

LOS, length of surgery; SRPU, surgery-related pressure ulcer

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Table 2 Characteristics of the included studies for dose–response meta-analysis for length of surgery and pressure ulcer risk

First author, year, country, study design, NOS SRPU definition (stage), SRPU total incidence	LOS category (min)	SRPU (+)	SRPU (–)	SRPU incidence	Odds ratio (95% CI)
Shen, 2015 (6)	<120	2	74	2·6%	1·0[ref.]
China	120-180	5	27	15·6%	6·851(1·025–74·392)
Retrospective cohort	180-240	13	65	16·7%	7·4(1·568–69·209)
8	240-300	12	42	22·2%	10·571(2·16–99·986)
I–IV	300-360	10	13	43·5%	28·462(4·98–280·784)
16·4%·	360-600	1	8	11·1%	4·625(0·07–95·572)
	>600	2	1	66·7%	74(2·277–4411·217)
Aronovitch, 1999 (17)	<180	26	423	5·8%	1·0[ref.]
USA	180-240	27	276	8·9%	1·592(0·873–2·903)
Retrospective survey	240-300	20	182	9·9%	1·788(0·92–3·422)
7	300-360	6	75	7·4%	1·302(0·423–3·377)
I–IV	360-420	3	34	8·1%	1·356(0·265–5·058)
8·1%	420-480	2	19	9·5%	1·713(0·184–7·743)
	>480	8	15	34·8%	8·677(2·883–24·059)

year; sample size of participants; age and gender of participants; SRPU definition (I–IV or II–IV); SRPU incidence; sample size, mean and standard deviation of LOS both in the SRPU group and the no SRPU group; LOS category; and number of patients in the case or control group in each LOS category.

The quality of the case–control and cohort studies was assessed according to the Newcastle–Ottawa Scale (NOS) (9). The quality of the cross-sectional study was also assessed by the NOS. The NOS contains eight items, categorised into three dimensions, including selection, comparability and exposure. The NOS ranges between zero and nine.

Statistical analysis

First, we compared LOS between the SRPU(+) group and SRPU(−) group of the included studies. The meta-analysis model was chosen by the heterogeneity between included studies. Statistical heterogeneity was explored by χ^2 and inconsistency (I^2) statistics; $P < 0{\cdot}05$ for χ^2 or I^2 value of 25% or more represented heterogeneity. In the absence of significant heterogeneity, the LOS of the included studies were pooled for weighted mean difference (WMD) using a fixed-effect model. If heterogeneity was observed, a random-effects model was used. Overall effects were determined using the Z test. Meta-regression was conducted for the source of heterogeneity by SRPU incidence. Publication bias was evaluated using a funnel graph, Egger's regression test and Begg's adjusted rank correlation test.

Second, we examined the dose–response relationship of LOS and pressure ulcer risk. The dose–response meta-analysis was carried out using the method proposed by Greenland and Longnecker (10) and Orsini *et al.* (11,12). Non-linear dose–response relationship was tested by using restricted cubic splines with three knots.

All statistical analyses were performed with Stata software, version 13·0 (Stata Corp, College Station, TX). Two-sided $P < 0{\cdot}050$ was considered statistically significant.

Results

Characteristics of included studies

We identified eight studies (6–8,13–17) that met our inclusion criteria for meta-analysis. The detailed steps of our literature search are shown in Figure 1. The studies included three prospective cohort studies, three retrospective cohorts and two cross-sectional surveys. Of these studies, three were conducted in USA, three in China, two in Japan, and Netherlands and Belgium contributed one study each. The quality rating of the included studies ranged from seven to eight on a scale of nine. Table 1 showed the characteristics of the eight identified studies.

LOS difference between the SRPU group and the no SRPU group

Six studies (7,8,13–16) with 976 patients were included in this comparison. The mean LOS in the SRPU (+) groups ranged from 252·5 to 335·7 minutes, compared with 233·0–298·3 minutes in the SRPU (−) groups. There was moderate heterogeneity between the included studies ($\chi^2 = 8{\cdot}38$, $P = 0{\cdot}137$; $I^2 = 40{\cdot}3\%$) (5). The pooled analysis using a random-effects model showed that LOS was higher in SRPU (+) groups compared with SRPU (−) groups (WMD = 36·081 minutes; 95% CI: 21·640–50·522 minutes; $Z = 4{\cdot}90$, $P = 0{\cdot}000$). Table 1 and Figure 2 showed the comparison of LOS between the two SRPU groups of the eight identified studies. The funnel plot showed no asymmetry, which indicated no evidence of publication bias (Figure 3). Begg's test ($z = 0{\cdot}38$, $P = 707$) and Egger's test ($t = 0{\cdot}60$, $P = 0{\cdot}583$) also suggested no publication bias. Meta-regression analysis showed that the heterogeneity did not come from SRPU incidence ($t = -0{\cdot}59$, $P = 0{\cdot}587$) (Figure 4).

Dose–response association between LOS and SRPU risk

Two studies (6,17) with 1254 patients were included in the dose–response meta-analysis (Table 2). A significant

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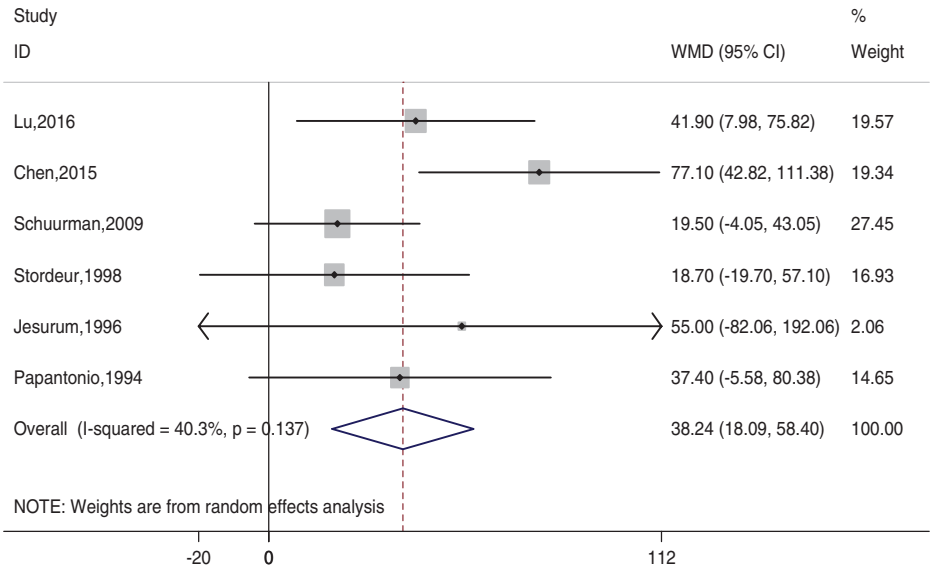

Figure 2 Comparison of length of surgery between SRPU (+) group and SRPU (–) group of the eight identified studies. Forest plot showed that length of surgery was higher in SRPU (+) groups.

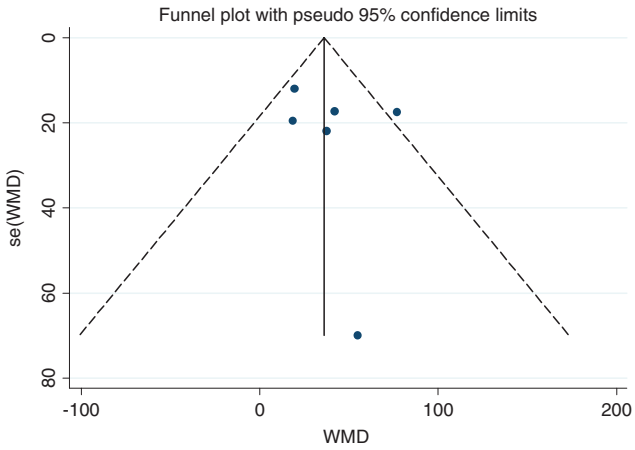

Figure 3 Publication bias analysis for length of surgery comparison between two groups. Funnel plot was asymmetry, which indicated no evidence of publication bias

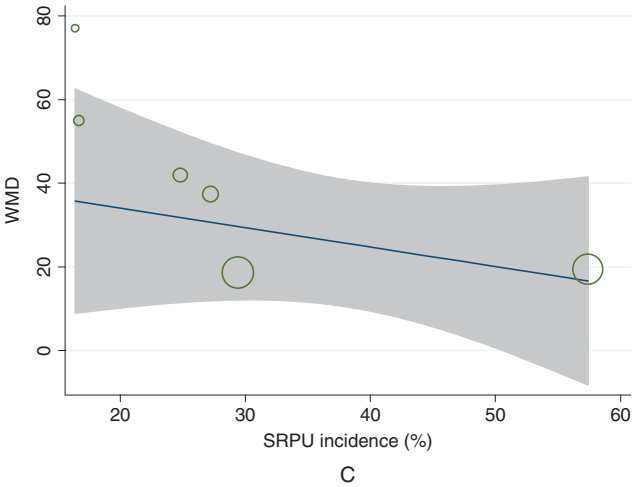

Figure 4 Meta-regression that showed heterogeneity was not obtained from SRPU incidence ($t = -0{\cdot}59$, $P = 0{\cdot}587$).

dose–response association was also found between LOS and risk of SRPU (model $\chi^2 = 9{\cdot}29$, $P = 0{\cdot}000$). In the linear model, the SRPU OR was 1·296 (95% CI 1·097–1·531) for a 60-minute increase in LOS intervals, 2·176 (95% CI 1·320–3·587) for a 180-minute increase, for a 300-minute increase and 13·344 (95% CI 2·521–70·636) for a 600-minute increase. In the spline model, the OR of SRPU increased almost linearly along with the LOS. Figure 5 shows the dose–response association between LOS and SRPU risk.

Discussion

Our meta-analysis indicated that SRPU risk increased almost linearly along with the LOS in cardiovascular surgical patients. The point estimate of LOS at 300 and 600 minutes increased the SRPU risk of 3·653 (95% CI 1·588–8·405) and 13·344 (95% CI 2·521–70·636), respectively, compared with the reference (60 minutes). In a pig model, Daniel found that pressure exceeding 35 mm Hg for 2 hours would cause ischemia resulting in a pressure ulcer (18). In rat models, Linder *et al.* found that even relatively short exposures (15 minutes to 1 hour) to pressures greater than 240 mm Hg would cause cell death; for exposures of 2 hours or greater, the magnitude of pressure is the important factor for causing cell death, and for the intermediate exposures (over 1 and 2 hours), the magnitude of cell death-causing pressure strongly depends on the time of exposure (19). In cardiac procedures, the LOS often exceeds 2 hours. The results of these animal experiments well explained why prolonged surgery leads to pressure ulcer and why LOS has a dose–response relationship with SRPU risk.

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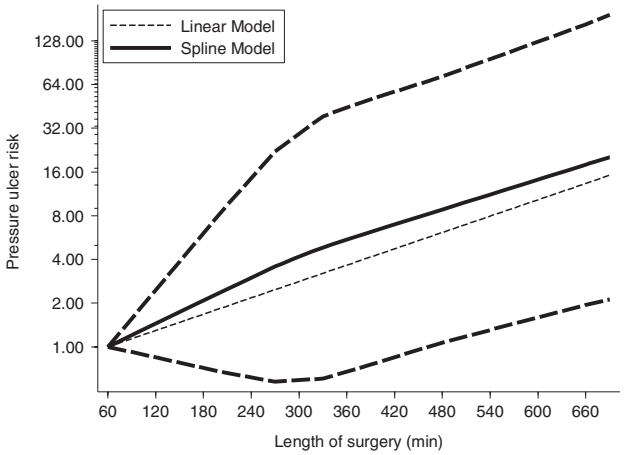

Figure 5 Dose–response relationships between length of surgery and pressure ulcers risk in cardiovascular surgical patients.

However, SRPUs in cardiovascular surgical patients not only result in prolonged pressure from operating table during cardiopulmonary bypass, hypothermia is often required for the purpose of organ protection, vasoactive drugs, hypotensive periods and reduced haemoglobin and haematocrit levels (5). This status mainly related to the cardiopulmonary bypass. Some studies also assessed the length of cardiopulmonary bypass and the risk of SRPUs in cardiovascular surgical patients. Lewicki reported that the mean extracorporeal circulation time for patients who developed pressure ulcers was 1·82 hours (SD = 0·84 hours) compared with 1·75 hours (SD = 0·67 hours) for patients who did not develop pressure ulcers (20). While Shen *et al.* reported that the length of cardiopulmonary bypass was not found to be statistically significantly different between the two groups [37 minutes (15–144 minutes) versus 44 minutes (16–107 minutes), $P = 0{\cdot}830$] (6). However, we did not assess the relationship between length of cardiopulmonary bypass and the risk of SRPU by meta-analysis because the existed studies did not provide sufficient data for this meta-analysis.

Some strategies have been used for the prevention SRPUs in cardiovascular surgery. A meta-analysis showed that patients who were provided a support surface had a significantly decreased incidence of surgery-related PUs (OR 0·31 [95% CI 0·17–0·59]) compared to patients using a standard mattress (21). Pham also reported that support surfaces for intraoperative prevention of pressure ulcers in patients undergoing surgery is cost-effective (22).Our meta-analysis found that LOS has dose–response relationship with SRPU risk in cardiovascular surgery patients. However, we still do not know what are the best prevention strategies stratified by LOS. It appears that well-designed prospective cohorts are needed.

There are some limitations in this meta-analysis. First, some of the included studies were retrospective case–control studies, which were more prone to bias. Second, there was moderate heterogeneity between the included studies. Meta-regression analysis of heterogeneity was not obtained from SRPU incidence. Heterogeneity may come from different study designs, ethnicity and/or SRPU stage. These two limitations need to be considered when evaluating the conclusion.

In conclusion, our meta-analysis indicated that LOS was an important risk factor for pressure ulcers in cardiovascular surgical patients. Some pressure ulcer prevention strategies should be carried out for the prolonged surgery for cardiovascular patients.

Acknowledgements

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